AIR FRYER MARKET ANALYSIS

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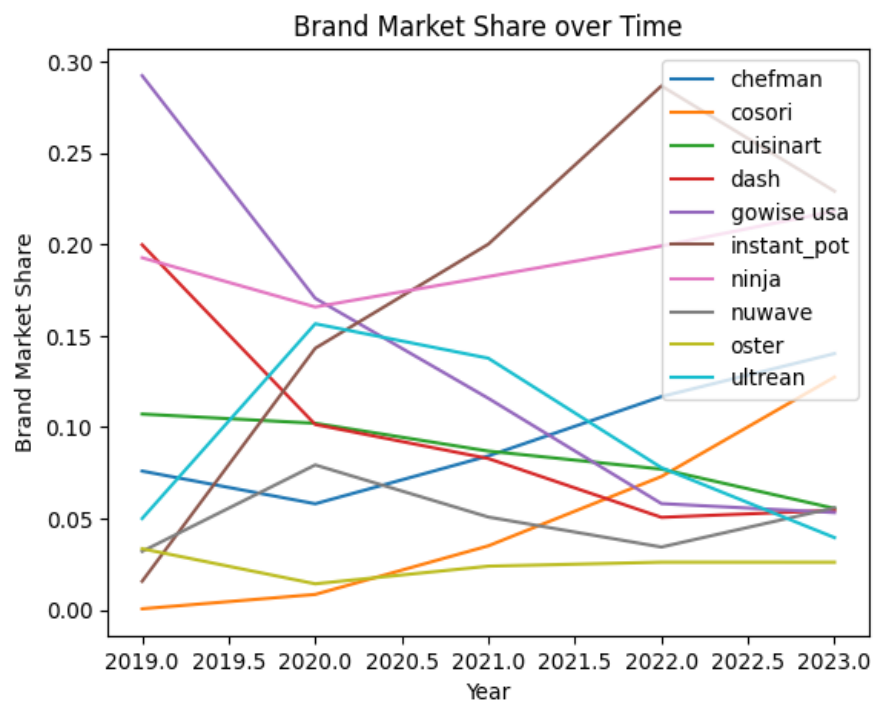
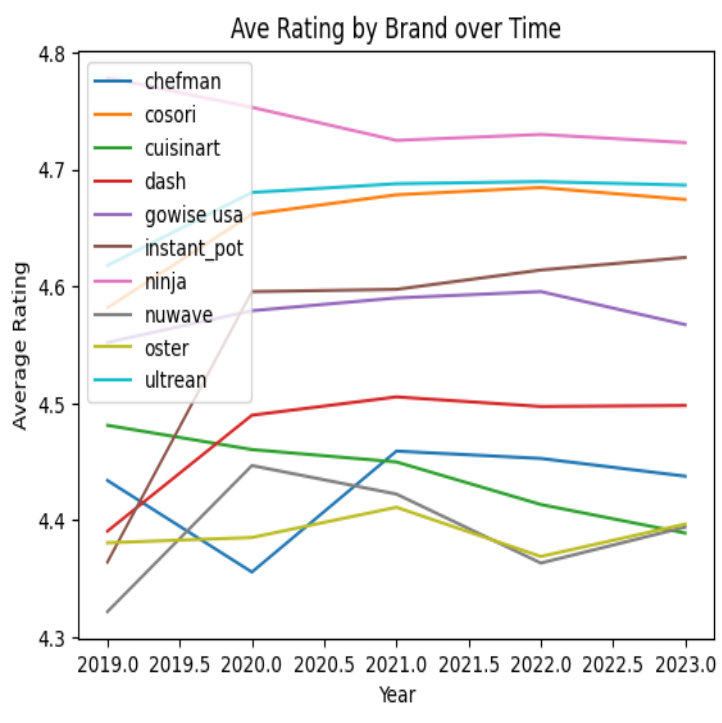
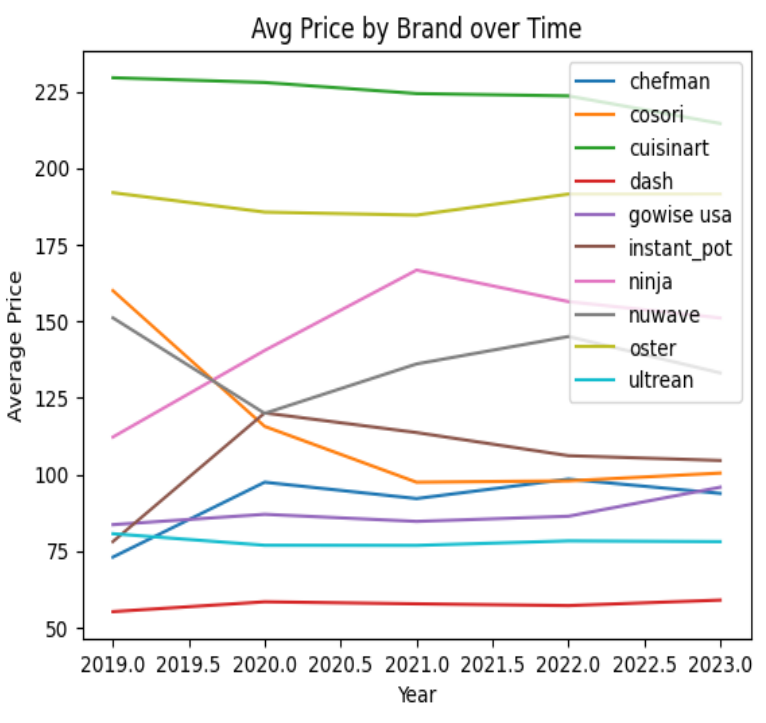
MARKET OVERVIEW

- Market is dominated by mass brands (Ninja, Instant Pot), with premium brands like Cuisinart and Oster occupying a smaller share due to high prices and low ratings
- Ninja dominates with strong ratings + share
- Explosive growth for Instant Pot by lowering prices and improving ratings
- High ratings does not mean low prices, with highest-rated Ninja being above all other highly-rated brands in price
- Appeal depends on more than just price







DEMAND MODEL

- We estimate demand using a multinomial logit model:

$$\log(sbt) = \alpha + \beta(\text{price } pbt) + \beta(\text{rating } rbt) + \text{features} + \text{brand dummies} + \text{year dummies}$$

- We model how market share depends on price, ratings, and product features
- Brand and year dummies are interpreted relative to the dropped reference categories (Chefman and 2019, respectively)

- **Estimated price coefficient:** β_price = -0.0377
- As expected, higher prices translate to lower demand
- Market share is price-sensitive

PRICE AND FEATURES

Top demand drivers (brands with these features tend to have greater market share):

1. Viewing windows
2. Compact designs
3. Oven-style models

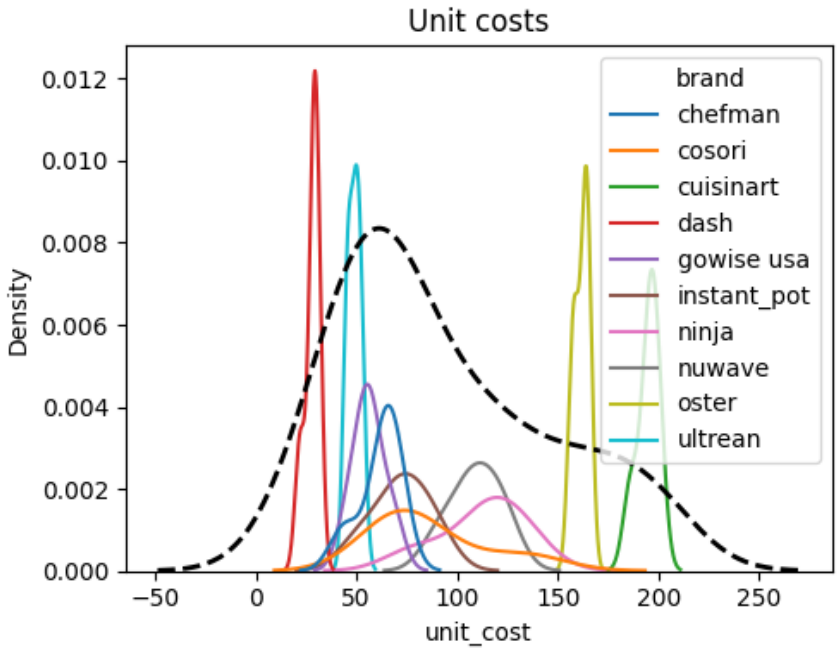
Compact and oven-style are both the most common features across brands and the top demand drivers, indicating that consumers commonly expect these features in an air fryer.

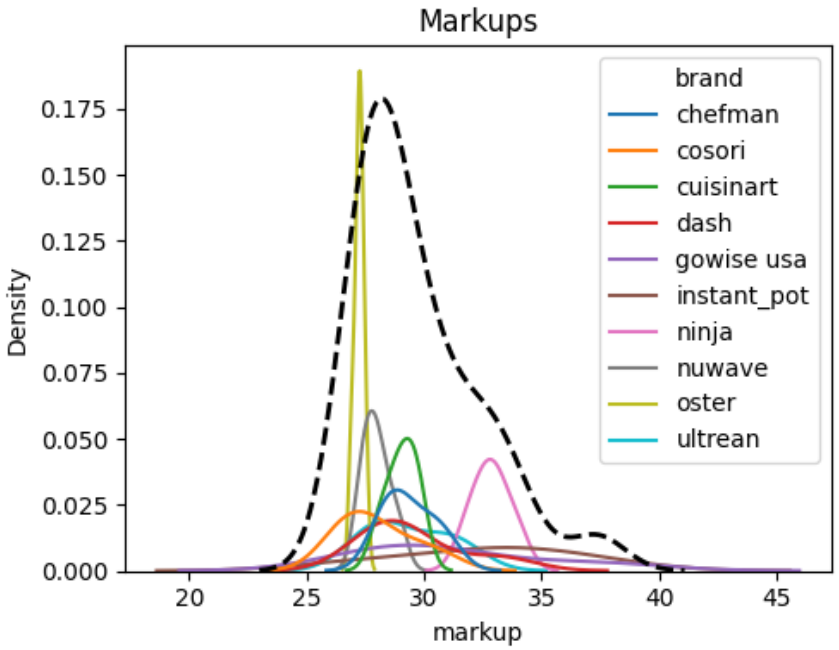
On the flipside, brands with dual-basket designs and rotisserie-capable models tend to have smaller market share.

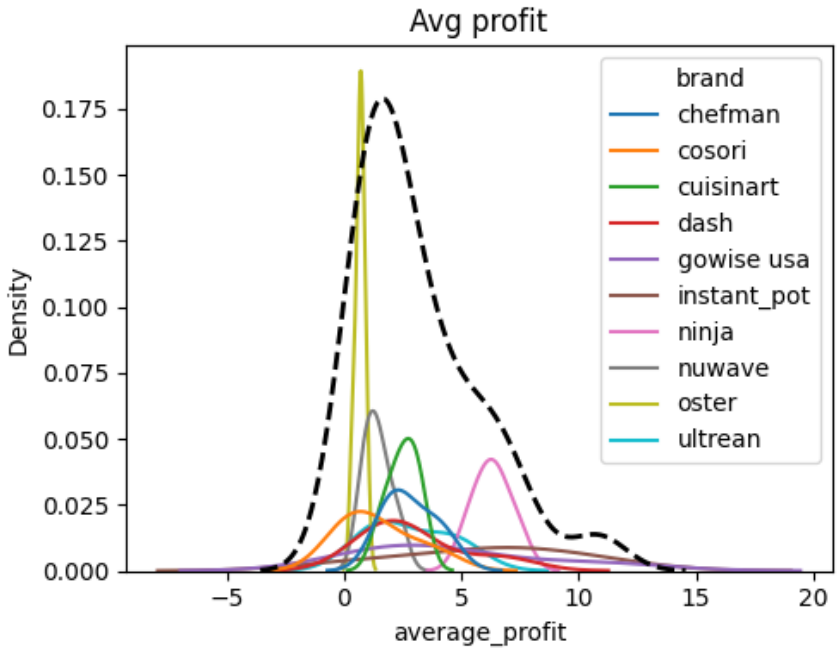
COSTS, MARKUPS, PROFITS

Using our demand model, we estimated unit costs, markups, and share-weighted profits:

- Premium brands show higher markups but lower market share
- Mass-market brands (e.g., Ninja) achieve higher share-weighted profits
- Cost and price differences reflect product positioning strategy







CONCLUSION

- Air fryer market is competitive and price-sensitive
- Mid-priced brands dominate demand, balancing quality and affordability
- Based on our model, a brand's winning strategy includes:
  - Competitive pricing
  - Strong consumer ratings
  - Practical, in-demand, high-value features

- Ninja occupies the strongest overall position
- Instant Pot exhibits the fastest growth trajectory
- Premium brands like Cuisinart and Oster face declining market share in the face of price decreases, rating drops, and profit losses